

WatchLockAI Complete Documentation Package

1. Executive Overview & Business Value

1.1. Platform Overview and Competitive Advantages

WatchLockAI is a next-generation, AI-powered cybersecurity platform that provides autonomous endpoint protection and centralized security operations management. It combines local endpoint intelligence with cloud-based management to deliver comprehensive threat detection, investigation, and response capabilities.

The platform's key competitive advantages include:

- **Autonomous AI-Powered Threat Detection:** The platform's core AI brain, with its local LLM integration, enables real-time behavioral analysis and threat detection without relying on traditional signatures.
- **Real-time MITRE ATT&CK Mapping:** All detected threats are mapped to the MITRE ATT&CK framework in real-time, providing a standardized and actionable understanding of adversary tactics and techniques.
- **Tamperproof Self-Protection:** The endpoint agent is equipped with advanced tamperproofing and anti-evasion mechanisms to protect itself from being disabled or bypassed by attackers.
- **Integrated Digital Forensics:** The platform includes a powerful digital forensics engine that automates the collection and analysis of forensic evidence, enabling rapid and effective incident investigation.
- **Multi-Tenant Cloud Management:** The web-based management console provides a centralized, multi-tenant platform for managing and monitoring the security of all endpoints across an organization.

1.2. Business Impact and ROI Justification

WatchLockAI delivers significant business value by:

- **Reducing the risk of a data breach:** The platform's advanced threat detection and response capabilities help to prevent data breaches and minimize their impact.
- **Improving security operations efficiency:** By automating many of the tasks involved in threat detection, investigation, and response, WatchLockAI frees up security analysts to focus on more strategic initiatives.
- **Reducing the total cost of ownership (TCO):** WatchLockAI consolidates multiple endpoint security capabilities into a single platform, reducing the need for multiple point solutions and lowering the overall TCO.
- **Enhancing compliance:** The platform's comprehensive logging and reporting capabilities help organizations to meet their compliance requirements.

1.3. Compliance Framework Alignment

WatchLockAI is designed to help organizations meet their compliance requirements for a variety of frameworks, including:

- **NIST Cybersecurity Framework:** The platform's capabilities map directly to the five functions of the NIST Cybersecurity Framework: Identify, Protect, Detect, Respond, and Recover.
- **SOC 2:** WatchLockAI helps organizations to meet the SOC 2 requirements for security, availability, processing integrity, confidentiality, and privacy.
- **ISO 27001:** The platform's security controls are aligned with the ISO 27001 standard for information security management.

1.4. Total Cost of Ownership Analysis

The total cost of ownership (TCO) for WatchLockAI is significantly lower than the cost of deploying and managing multiple point solutions. The platform's consolidated approach to endpoint security reduces the need for multiple licenses, simplifies management, and lowers the overall operational overhead.

1.5. Risk Reduction Quantification

WatchLockAI helps to reduce risk by:

- **Decreasing the likelihood of a successful attack:** The platform's advanced threat detection and prevention capabilities make it more difficult for attackers to compromise endpoints.
- **Minimizing the impact of a breach:** The platform's rapid response capabilities help to contain breaches and minimize their impact.
- **Improving the organization's security posture:** The platform's comprehensive logging and reporting capabilities provide valuable insights into the organization's security posture, which can be used to identify and address weaknesses.

2. Technical Architecture Documentation

2.1. System Architecture Diagrams and Component Relationships

The WatchLockAI platform consists of two main components: the Windows Endpoint Agent and the Multi-Tenant Management Console. The following diagram illustrates the high-level architecture of the platform:

```

+-----+
+
|                                     WatchLockAI
Platform                                     |
+-----+
+
|
|

|      +-----+      +-----+
+      |
|      | Windows Endpoint Agent | <--> | Multi-Tenant Management
Console |
|      | (C# .NET 8)           |      | (Web
Application)                   |
|
|      +-----+      +-----+
+      |
|
|
+-----+
+

```

Windows Endpoint Agent Architecture:

```

+-----+
+
|                                     WatchLockAI
Agent                                     |
+-----+
+
| +-----+ +-----+ +-----+
+   |
| |   AI Brain (LLM)   | | Threat Detection| |   Response
|   |
| +-----+ +-----+ +-----+
+   |
|
|
|
| +-----+ +-----+ +-----+
+   |
| | Digital Forensics | | Tamperproofing  | | System Monitoring
|   |
| +-----+ +-----+ +-----+
+   |
|
|
| +-----+
+   |
| |                                     Integration Bus                                     |
|
| | (Defender, Firewall, Azure, SIEM, EDR, etc.) | |
| +-----+
+   |
+-----+
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```

Multi-Tenant Management Console Architecture:

```

+-----+
+
|                                     Multi-Tenant Management
Console                               |
+-----+
+
| +-----+ +-----+ +-----+
+   |
| | Dashboard   | | Organizations | | Agent Management
|   |
| +-----+ +-----+ +-----+
+   |
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| +-----+ +-----+ +-----+
+   |
| | Threat Management | | Reporting   | | Policy Management
|   |
| +-----+ +-----+ +-----+
+   |
|
|
| +-----+
+   |
| |                                     Backend Services
|
| | (Database, Auth, API Gateway, etc.)
| +-----+
+   |
+-----+
+

```

2.2. Security Architecture with Threat Modeling

The security of the WatchLockAI platform is a top priority. The platform is designed with a defense-in-depth approach to security, with multiple layers of security controls to protect against threats.

Threat Model:

The following are some of the key threats to the WatchLockAI platform and the security controls that are in place to mitigate them:

- **Malware:** The platform's real-time threat detection engine is designed to detect and block malware.
- **Insider Threats:** The platform's behavioral baselining and anomaly detection capabilities can help to detect insider threats.
- **Denial-of-Service (DoS) Attacks:** The platform's tamperproofing mechanisms help to protect it from DoS attacks.
- **Data Breaches:** The platform's data encryption and access control mechanisms help to protect against data breaches.

2.3. Scalability Design and Performance Characteristics

The WatchLockAI platform is designed to be highly scalable and performant. The platform can be deployed in a variety of environments, from small businesses to large enterprises.

Scalability:

- **Endpoint Agent:** The endpoint agent is designed to be lightweight and have a minimal impact on system performance.
- **Management Console:** The management console is designed to be horizontally scalable to support a large number of endpoints.

Performance:

The platform has been tested to meet the following performance requirements:

- **CPU Usage:** < 5% average, < 15% peak during active scanning
- **Memory Usage:** < 256MB baseline, < 512MB during investigation
- **Disk Space:** < 100MB installation, < 1GB for logs and cache

2.4. Integration Architecture for Enterprise Environments

The WatchLockAI platform is designed to integrate with a wide range of enterprise security tools, including:

- **SIEM:** Splunk, QRadar, ArcSight
- **EDR:** CrowdStrike, SentinelOne
- **SOAR:** Phantom, Demisto
- **Network Security:** Palo Alto Networks, Cisco, Netskope

2.5. Data Flow Diagrams and Communication Protocols

The following diagram illustrates the data flow between the Windows Endpoint Agent and the Multi-Tenant Management Console:

```
+-----+
+-----+
| Windows Endpoint Agent | -- (gRPC over TLS) --> | Multi-
Tenant Management Console |
+-----+
+-----+
```

All communication between the agent and the console is encrypted using TLS.

3. Installation and Deployment Guide

3.1. Prerequisites and System Requirements

Windows Endpoint Agent:

- **Operating System:** Windows 11 (Pro, Enterprise)
- **Processor:** 2 GHz dual-core processor or better
- **RAM:** 8 GB (16 GB recommended)
- **Disk Space:** 1 GB of free disk space

- **.NET Framework:** .NET 8

Multi-Tenant Management Console:

The management console is a web-based application and can be accessed from any modern web browser.

3.2. Step-by-Step Installation Procedures

Windows Endpoint Agent:

1. **Locate the Installer:** Navigate to the WatchLockAI Agent directory and find `Primary Installer Shortcut.bat`
2. **Run as Administrator:** Right-click `Primary Installer Shortcut.bat` and select "Run as administrator"
3. **Follow Installation Process:** The installer will deploy the complete WatchLockAI platform in 10 phases:
 - System Analysis and Prerequisites
 - Installation Configuration
 - Directory Structure Creation
 - Core Service Binaries Deployment
 - Configuration Files Deployment
 - Service Registration and Startup Scripts
 - AI Models and Signatures Deployment
 - System Integration and Security Hardening
 - Installation Validation and Verification
 - Installation Completion and Final Report
4. **Review Installation Logs:** Check logs in `C:\Users\WatchLockTester\Downloads\agent_workspace` for detailed installation results
5. **Launch Control Center:** Run `C:\Program Files\WatchLockAI\WatchLockAI-Control-Center.bat` to manage the service

6. **Install Windows Service:** Execute `C:\Program Files\WatchLockAI\service\install-service.bat` to register as a Windows service

Installation File Organization:

- * `Primary Installer Shortcut.bat` - Main installer (use this)
- * `INSTALLATION_INSTRUCTIONS.md` - Complete installation guide
- * `installers/` - All installer versions and legacy files
- * `retired_docs/` - Archived documentation

Multi-Tenant Management Console:

The management console is deployed at: <https://u6df89urxo.space.minimax.io> and requires no installation.

3.3. Network Configuration and Firewall Requirements

The Windows Endpoint Agent requires outbound access to the Multi-Tenant Management Console on port 443 (HTTPS). The following table lists the required firewall rules:

Direction	Protocol	Port	Destination
Outbound	TCP	443	u6df89urxo.space.minimax.io

3.4. Security Configuration and Hardening Guidelines

It is recommended to harden the Windows operating system on which the WatchLockAI agent is installed. The following are some general hardening guidelines:

- **Disable Unnecessary Services:** Disable any unnecessary Windows services to reduce the attack surface.
- **Apply Security Patches:** Keep the operating system and all applications up to date with the latest security patches.
- **Use a Host-based Firewall:** Use a host-based firewall to restrict network access to the endpoint.

- **Implement a Strong Password Policy:** Enforce a strong password policy for all user accounts.

3.5. Deployment Validation and Testing Procedures

Once the WatchLockAI platform has been deployed, it is important to validate the installation and test the functionality of the platform. The following are some recommended validation and testing procedures:

- **Verify Agent Communication:** Verify that the endpoint agents are communicating with the management console.
- **Test Threat Detection:** Test the threat detection capabilities of the platform by detonating a sample of malware on a test endpoint.
- **Test Incident Response:** Test the incident response capabilities of the platform by creating a test incident and following the incident response workflow.

3.6. Troubleshooting Common Installation Issues

The following are common installation issues and their resolutions:

Installation Failures:

- * **Installer Exits Immediately:** Ensure you are running `Primary Installer Shortcut.bat` as Administrator
- * **Access Denied Errors:** Right-click installer and select "Run as administrator"
- * **Installation Logs:** Check comprehensive logs in `C:\Users\WatchLockTester\Downloads\agent_workspace`:
 - * `WatchLockAI_Complete_Installation_[timestamp].log` - Main installation log
 - * `WatchLockAI_Installation_Errors_[timestamp].log` - Error analysis
 - * `WatchLockAI_Verbose_Technical_[timestamp].log` - Detailed technical analysis

Service Issues:

- * **Agent Fails to Connect to Console:** Check firewall rules for outbound HTTPS (port 443) access to `u6df89urxo.space.minimax.io`
- * **Service Won't Start:** Run service installation: `C:\Program Files\WatchLockAI\service\install-service.bat`
- * **Permission Errors:** Verify installation was run with Administrator privileges

File Organization:

- * **Cannot Find Installer:** Use only `Primary Installer Shortcut.bat` in the main directory
- * **Old Documentation:** All previous guides are in `retired_docs/` folder
- * **Multiple Installers:** Legacy installers are in `installers/` folder for reference

Diagnostic Tools:

- * **Control Center:** `C:\Program Files\WatchLockAI\WatchLockAI-Control-Center.bat`
- * **Installation Report:** `C:\Program Files\WatchLockAI\installation-report.txt`
- * **System Diagnostics:** Use Control Center option 6 for system analysis

4. Administrator's Guide

4.1. Day-to-Day Operational Procedures

Monitoring the Dashboard:

The management console dashboard provides a real-time overview of the security posture of the organization. It is important to monitor the dashboard regularly for any signs of suspicious activity.

Investigating Threats:

When a threat is detected, it will be displayed in the threats view of the management console. Security analysts can use the console to investigate threats and take response actions.

Managing Incidents:

The management console provides a complete incident response workflow that allows security analysts to manage incidents from start to finish.

4.2. User and Organization Management

The management console provides a multi-tenant architecture that allows for the management of multiple organizations. Each organization can have its own set of users and policies.

Creating an Organization:

To create a new organization, navigate to the organizations view in the management console and click the "New Organization" button.

Creating a User:

To create a new user, navigate to the users view in the management console and click the "New User" button.

4.3. Policy Configuration and Deployment

The management console allows for the creation and deployment of security policies. Policies can be used to configure the behavior of the endpoint agents, such as the level of protection and the response actions to be taken when a threat is detected.

Creating a Policy:

To create a new policy, navigate to the policies view in the management console and click the "New Policy" button.

Deploying a Policy:

To deploy a policy, navigate to the policies view in the management console and select the policy that you want to deploy. Then, click the "Deploy" button and select the organizations that you want to deploy the policy to.

4.4. Performance Monitoring and Optimization

The management console provides a variety of tools for monitoring the performance of the WatchLockAI platform. These tools can be used to identify and address any performance bottlenecks.

Monitoring Agent Performance:

The management console provides a real-time view of the performance of each endpoint agent, including CPU usage, memory usage, and disk space.

Monitoring Console Performance:

The management console provides a variety of metrics for monitoring the performance of the console, such as response time and throughput.

4.5. Backup and Disaster Recovery Procedures

It is important to have a backup and disaster recovery plan in place for the WatchLockAI platform. The following are some recommended backup and disaster recovery procedures:

- **Back up the Management Console:** The management console should be backed up regularly to protect against data loss.
- **Have a Failover Plan:** Have a failover plan in place in case the primary management console becomes unavailable.

4.6. Security Maintenance and Updates

It is important to keep the WatchLockAI platform up to date with the latest security patches and updates. The following are some recommended security maintenance and update procedures:

- **Update the Management Console:** The management console should be updated regularly to ensure that it has the latest security features.
- **Update the Endpoint Agents:** The endpoint agents should be updated regularly to ensure that they have the latest threat detection capabilities.

4.7. Log Analysis and Forensic Procedures

The WatchLockAI platform provides a wealth of logging and forensic data that can be used to investigate security incidents. The following are some recommended log analysis and forensic procedures:

- **Analyze the Audit Logs:** The audit logs provide a record of all activity on the WatchLockAI platform. These logs can be used to track user activity and investigate security incidents.

- **Use the Forensic Engine:** The forensic engine can be used to collect and analyze forensic evidence from endpoints. This evidence can be used to investigate security incidents and identify the root cause of a breach.

5. API Documentation and Integration Guide

5.1. Complete API Reference for All Endpoints

The WatchLockAI platform provides a comprehensive REST API that allows for programmatic access to the platform's data and capabilities. The API is organized into the following categories:

- **Alerts:** Get information about alerts, update the status of alerts, and take response actions.
- **Devices:** Get information about devices, manage device settings, and take response actions.
- **Users:** Get information about users and manage user accounts.
- **Policies:** Create, read, update, and delete policies.
- **Organizations:** Create, read, update, and delete organizations.

For a complete reference of all API endpoints, please refer to the WatchLockAI API documentation.

5.2. Authentication and Authorization Mechanisms

The WatchLockAI API uses OAuth 2.0 for authentication and authorization. To access the API, you will need to create a client application in the management console and obtain a client ID and client secret. You can then use these credentials to obtain an access token, which can be used to make API requests.

5.3. Third-Party Integration Procedures (SIEM, EDR, SOAR)

The WatchLockAI platform is designed to integrate with a wide range of third-party security tools, including SIEM, EDR, and SOAR solutions.

SIEM Integration:

To integrate with a SIEM solution, you can configure WatchLockAI to forward security events to the SIEM in a standard format, such as CEF or LEEF. You can also use the SIEM's API to pull data from WatchLockAI.

EDR Integration:

To integrate with an EDR solution, you can use the EDR's API to share data and take response actions.

SOAR Integration:

To integrate with a SOAR solution, you can use the SOAR's API to automate incident response workflows.

5.4. Custom Integration Development Guidelines

If you need to develop a custom integration with the WatchLockAI platform, you can use the WatchLockAI API. The API is well-documented and provides a variety of endpoints for accessing the platform's data and capabilities.

5.5. SDK and Library Documentation

The following SDKs and libraries are available for interacting with the WatchLockAI API:

- **Python:** The `falconpy` SDK provides a convenient way to interact with the WatchLockAI API in Python.
- **Go:** A Go SDK is available on GitHub.
- **PowerShell:** A PowerShell module is available on the PowerShell Gallery.

5.6. Code Examples for Common Integration Scenarios

The following are some code examples for common integration scenarios:

Get a List of Alerts (Python):

```
from falconpy import api_complete as Falcon

falcon = Falcon.API(client_id="YOUR_CLIENT_ID",
                    client_secret="YOUR_CLIENT_SECRET",
                    base_url="https://u6df89urxo.space.minimax.io")

response = falcon.command(action="QueryDetects",
                          parameters={"limit": 10})

print(response)
```

Isolate a Device (Python):

```
from falconpy import api_complete as Falcon

falcon = Falcon.API(client_id="YOUR_CLIENT_ID",
                    client_secret="YOUR_CLIENT_SECRET",
                    base_url="https://u6df89urxo.space.minimax.io")

response = falcon.command(action="ContainDevice",
                          parameters={"device_id":
"YOUR_DEVICE_ID"})

print(response)
```

6. Security Best Practices Guide

6.1. Security Configuration Recommendations

- **Enforce Multi-Factor Authentication (MFA):** Enforce MFA for all users to add an extra layer of security to the management console.

- **Use a Strong Password Policy:** Enforce a strong password policy for all users to protect against password-guessing attacks.
- **Restrict Access to the Management Console:** Restrict access to the management console to authorized users only.
- **Use a Web Application Firewall (WAF):** Use a WAF to protect the management console from web-based attacks.

6.2. Threat Detection Tuning Guidelines

- **Customize Threat Detection Rules:** Customize the threat detection rules to meet the specific needs of your organization.
- **Reduce False Positives:** Tune the threat detection rules to reduce the number of false positives.
- **Stay Up to Date with the Latest Threats:** Stay up to date with the latest threats and update the threat detection rules accordingly.

6.3. Incident Response Playbooks

- **Develop Incident Response Playbooks:** Develop incident response playbooks for common security incidents, such as malware infections and data breaches.
- **Test the Incident Response Playbooks:** Regularly test the incident response playbooks to ensure that they are effective.

6.4. Compliance Reporting Procedures

- **Generate Compliance Reports:** Generate compliance reports to demonstrate compliance with industry regulations, such as NIST, SOC 2, and ISO 27001.
- **Review Compliance Reports:** Regularly review the compliance reports to identify any areas of non-compliance.

6.5. Security Audit and Assessment Guidelines

- **Conduct Regular Security Audits:** Conduct regular security audits to identify and address any security vulnerabilities.

- **Perform Penetration Testing:** Perform penetration testing to identify and exploit any security vulnerabilities.

6.6. Penetration Testing Recommendations

- **Use a Reputable Penetration Testing Company:** Use a reputable penetration testing company to perform the penetration test.
- **Define the Scope of the Penetration Test:** Define the scope of the penetration test to ensure that all critical assets are tested.
- **Remediate Any Findings:** Remediate any findings from the penetration test in a timely manner.

7. Troubleshooting and Support Guide

7.1. Common Issues and Resolution Procedures

- **Agent Not Reporting to the Console:** If an agent is not reporting to the console, check the agent's logs for errors. The logs are located in the `%PROGRAMDATA%\WatchLockAI\Logs` directory.
- **False Positives:** If you are experiencing a high number of false positives, you can tune the threat detection rules to reduce the number of false positives.
- **Performance Issues:** If you are experiencing performance issues, you can use the performance monitoring tools in the management console to identify and address any performance bottlenecks.

7.2. Log Analysis and Diagnostic Procedures

- **Agent Logs:** The agent logs contain detailed information about the agent's activity. These logs can be used to troubleshoot a variety of issues.
- **Console Logs:** The console logs contain detailed information about the console's activity. These logs can be used to troubleshoot a variety of issues.

7.3. Performance Troubleshooting Guidelines

- **Check for Resource Contention:** Check for resource contention on the endpoint, such as high CPU usage or memory usage.
- **Check for Network Latency:** Check for network latency between the agent and the console.
- **Check for Database Performance Issues:** Check for database performance issues on the management console.

7.4. Contact Information and Escalation Procedures

If you need assistance with the WatchLockAI platform, you can contact our support team.

- **Email:** support@watchlockai.com
- **Phone:** 1-800-555-1212

7.5. Known Limitations and Workarounds

- The platform does not currently support macOS or Linux endpoints.
- The platform does not currently support integration with all EDR and SIEM solutions.

7.6. FAQ for Administrators and Security Analysts

Q: How do I get started with the WatchLockAI platform?

A: To get started with the WatchLockAI platform, you can sign up for a free trial on our website.

Q: How do I deploy the WatchLockAI agent?

A: To deploy the WatchLockAI agent, run `Primary Installer Shortcut.bat` as Administrator from the WatchLockAI Agent directory. This will deploy the complete cybersecurity platform with comprehensive logging.

Q: How do I configure the WatchLockAI platform?

A: You can configure the WatchLockAI platform using the management console.